

EXPERIENCE

ENARXI INNOVATION PVT.LTD. | AI/ML Engineer

Feb 2026 - Present | Chennai, India

- Developed an AI-powered **Conveyor Belt Monitoring** System for **JSW Steel** using **YOLOv8**, achieving **96%** defect detection accuracy for real-time predictive maintenance.
- Built a **Material Detection & Classification** pipeline processing **30+ FPS** with **95%** classification accuracy.
- Designed a **Cement Bag Counting** System using **Object Detection** and **Tracking**, achieving **99%** counting accuracy for automated inventory verification.
- Engineered a **BOM Automation** solution using **AI** reducing manual processing by **60%** while achieving **98%** data extraction accuracy.
- Developed an enterprise **AI Agent** using **LLMs**, **RAG**, **FastAPI**, and **Vector Databases**, reducing document retrieval time from minutes to seconds.

XTROP RESEARCH SOLUTIONS | AI/ML Research Intern

Jun 2025 - Aug 2025 | Chennai, India

- AI-powered **gallstone prediction** system using **ML**, **Genetic Algorithm**-optimized stacking **ensemble learning** on structured clinical data, delivering accurate and reliable clinical decision support.
- Deep learning**-based **tuberculosis detection** using **U-Net** for precise lung infection segmentation from chest X-ray images, enabling early and reliable diagnosis.

EDUCATION

Bachelor of Technology in Artificial Intelligence and Data Science

BS Abdur Rahman Crescent Institute Of Science and Technology

2022-2026 | Tamil Nadu, India

CGPA: 8.2

PROJECTS

Chatbot with RAG and Session Memory:

LangGraph-based AI chatbot using **Retrieval-Augmented Generation (RAG)** and session memory to provide context-aware responses through intelligent query routing and vector-based retrieval.

Task Manager AI:

AI-powered **task management** application that converts natural language into structured tasks using **NLP** and **Generative AI**, with **REST APIs** for task extraction, workflow management, and **JSON**-based data storage.

PUBLICATION

GallIntel- Intelligent Gallstone Detection using Genetically Optimized Stacking Model. [Published]

Developed an AI-powered gallstone detection system using a Genetic Algorithm-optimized stacking ensemble model for accurate prediction and clinical decision support.

TB-Xnet- Interpretable Deep Segmentation Model for Tuberculosis Detection [Published]

Developed an ML model for tuberculosis prediction using ensemble learning and optimized feature selection.

SKILLS

Programming Languages

- Python

Databases

- PostgreSQL
- SQL
- MongoDB

AI/ML

- Machine Learning
- Deep Learning
- Computer Vision
- NLP
- GenAI
- Image Processing
- LLM
- RAG

Frameworks & Tools

- LangChain
- LangGraph
- OpenCV
- YOLOv8
- FastAPI
- PyTorch

Automation & Development

- AI Automation
- AI Agent
- Chatbot
- Playwright

CORE COMPETENCIES

- Problem solving
- Analytical Thinking
- Communication Skills
- Teamwork & Collaboration
- Active Listening
- Time Management

CERTIFICATIONS

IEEE ICMSCI 2026 – Paper Presentation Certificate

IEEE ICERECT 2025 – Paper Presentation Certificate

Besant Technologies - Data Science

KEY ACHIEVEMENTS

AI / Machine Learning Project Achievement

Built and evaluated AI/ML models for real- world problem solving using machine learning and deep learning techniques.

AI-powered application

Developed an AI-powered application using Python, Large Language Models (LLMs), and Retrieval-Augmented Generation (RAG) with database integration for intelligent task automation and contextual responses.